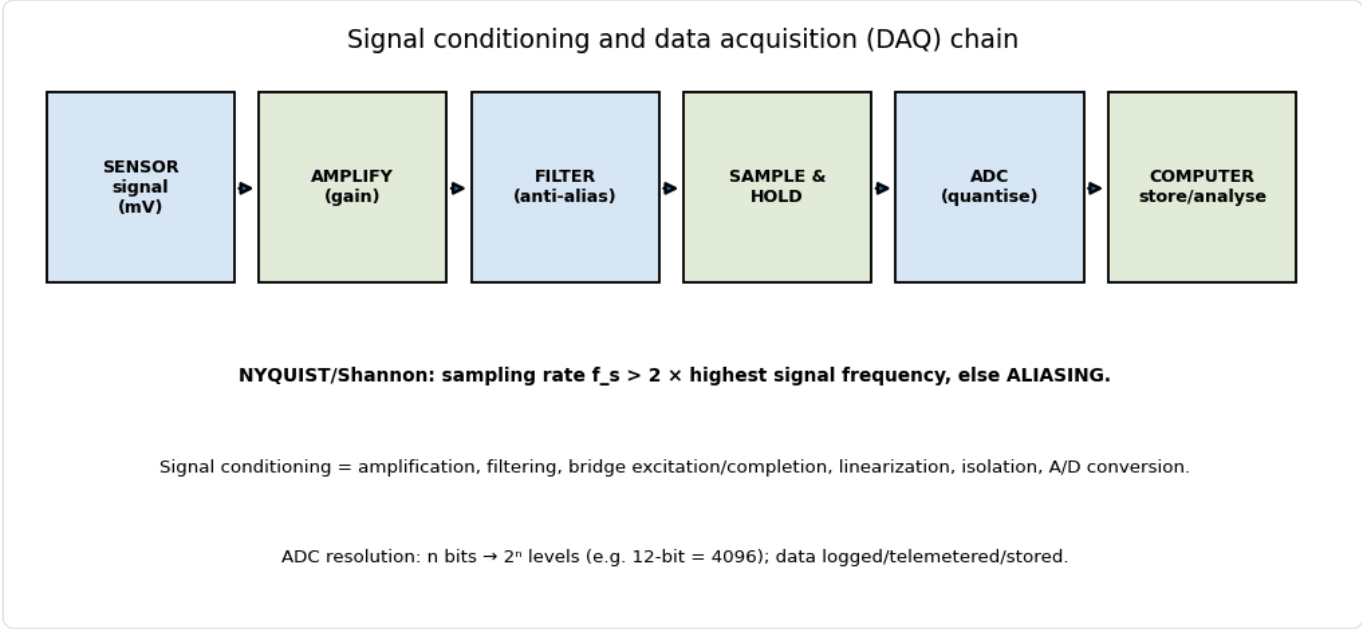
UNIT 9 · TOPIC 3

Signal Conditioning and Transmission

Syllabus topic: Signal transmission, processing, conditioning and monitoring



Signal conditioning and data acquisition (DAQ) chain

Signal conditioning

The raw transducer signal is often only millivolts. Conditioning prepares it for recording:

- Bridge excitation and completion.
- AMPLIFICATION (instrumentation amplifier).
- FILTERING — remove noise; an anti-alias filter before sampling (Topic 4).
- Linearization and isolation.
- Modulation/demodulation for AC methods.

Signal transmission

Method	Use
Voltage signal	Short runs
4–20 mA CURRENT loop	Long distances — noise-immune
Digital	Networked instruments
SLIP RINGS or telemetry	Signals from rotating shafts
Wireless / IoT telemetry	Now common (Unit 2, Topic 10)

Practice MCQs

Q24. Signal conditioning does NOT include:

(a) Amplification
(b) Filtering
(c) Casting
(d) Linearization

Q25. A 4–20 mA current loop is used for signal transmission because it is:

(a) Cheaper only
(b) Noise-immune over long distances
(c) Digital
(d) Wireless

Q26. Signals from a rotating shaft are carried out by:

(a) Slip rings or telemetry
(b) Gears
(c) Belts
(d) Chains

Answers

Q24 — (c) Casting

Q25 — (b) Noise-immune over long distances

Q26 — (a) Slip rings or telemetry

Sources: Doebelin — Measurement Systems; Beckwith — Mechanical Measurements; Sawhney — Electrical & Electronic Measurements; HBM/Omega strain-gauge technical notes (as given in the source notes, July 2026); ASAE instrumentation standards; NPTEL measurement lectures.